

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456240.5922 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.046
Transit depth (Rp/Rs)^2	0.98 +/- 0.91 [%]
Orbital Inclination (inc)	89.34 +/- 2.91
Airmass coefficient 1 (a1)	3.28 +/- 1.83
Airmass coefficient 2 (a2)	-0.78 +/- 0.63
Scatter in the residuals of the lightcurve fit is	51.99 %
Best Comparison Star	#2 - [551, 426]
Optimal Aperture	1.7
Optimal Annulus	7.02
Transit Duration (day)	0.0554 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.099 ± 0.046 vs 0.1326 (deviation 0.53σ)
Duration fit vs published	0.0554 d vs 0.0737 d (deviation 1.59σ)
Mid-time O-C	7.0 min
Depth SNR	1.6
Residual scatter	51.99 %

Light curve

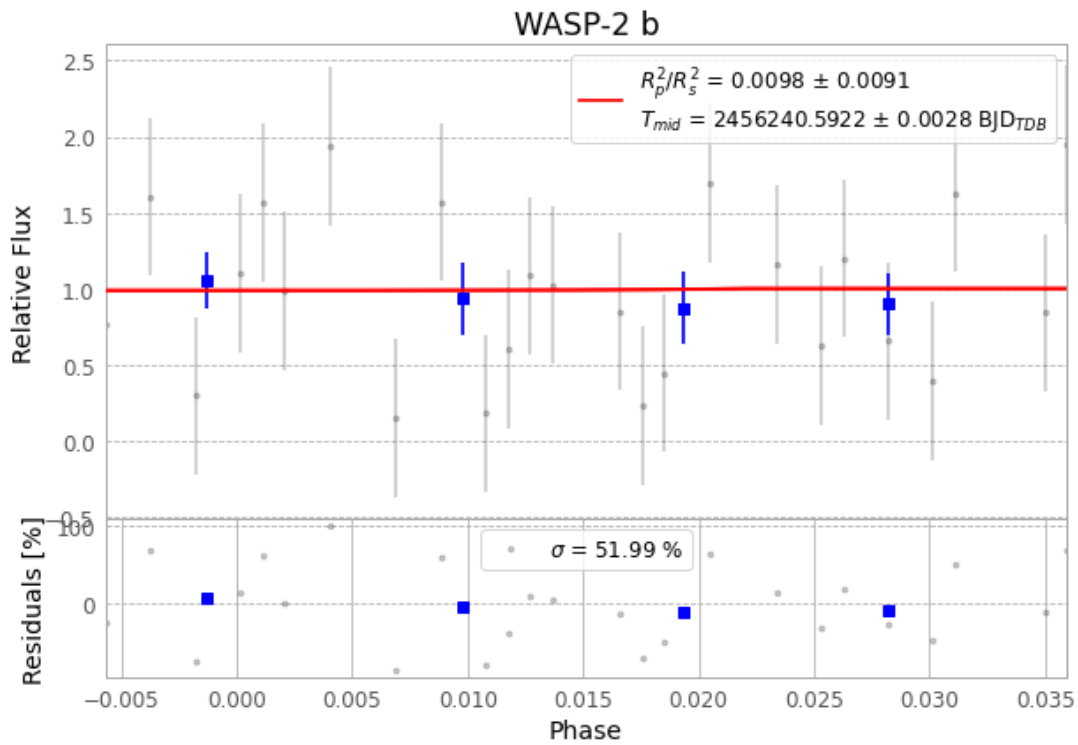


Figure 1 — FinalLightCurve_WASP-2 b_2012-11-08.png (SHA-256 967db6b22112be1b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [518, 303], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 585984f015549e9c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

47 FITS frames (31 MB), WASP-2121109014813.FITS → WASP-2121109040612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-121109.log (e5551dbfd1a2...), FinalLightCurve_WASP-2 b_2012-11-08.png (967db6b22112...), FinalLightCurve_WASP-2 b_2012-11-08.csv (abc0ada37646...)

Compute resources

Wall time 111.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:44Z.